SUPPORTING ADULT GUIDANCE

Surds: manipulating and rationalising

Number Systems and Calculator Judgement · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson extends surd work to expanding brackets and rationalising denominators. Learners multiply surds, then use a 2 by 2 expansion grid (as in Quadratics) to expand brackets; in conjugate grids the surd cells visibly cancel. They then rationalise denominators step by step — choosing the multiplier, finding the new denominator, simplifying — and explore which conjugate pairs give products of 1, 2 or a negative number. A stretch proves $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$.

Driving Question: *How do I tidy a surd out of the denominator of a fraction?*

The mathematics, in plain terms

For $b \geq 0$, $\sqrt{b} \times \sqrt{b} = b$. Expanding $(a + \sqrt{b})(a - \sqrt{b})$ gives $a^2 - a\sqrt{b} + a\sqrt{b} - b = a^2 - b$, the difference of two squares. Rationalising multiplies a fraction by a form of 1 ($\sqrt{b}/\sqrt{b}$ or the conjugate over itself), so its value is unchanged while the denominator becomes rational. Historically this made hand division practical; today it gives a standard form that is easy to compare and combine.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The expansion grid

Expand $(3 + \sqrt{2})(3 - \sqrt{2})$.

$(3 + \sqrt{2})(3 - \sqrt{2})$

×	**3**	$-\sqrt{2}$
3	9	$-3\sqrt{2}$
$\sqrt{2}$	$3\sqrt{2}$	-2

1. Fill each cell: its row term × its column term.	$\sqrt{2} \times (-\sqrt{2}) = -2$
2. The shaded surd cells, $-3\sqrt{2}$ and $+3\sqrt{2}$, are equal and opposite.	they cancel
3. Collect the whole numbers.	$9 - 2 = 7$

$(3 + \sqrt{2})(3 - \sqrt{2}) = 7$. No surd is left: these brackets are conjugates.

Example 2 — Rationalising with the conjugate

Rationalise the denominator of $4/(3 + \sqrt{5})$.

The new denominator: $(3 + \sqrt{5})(3 - \sqrt{5})$

×	**3**	$-\sqrt{5}$
3	9	$-3\sqrt{5}$
$\sqrt{5}$	$3\sqrt{5}$	-5

1. Multiply the numerator and the denominator by $3 - \sqrt{5}$.	this is multiplying by 1
2. Denominator: the grid's surd cells cancel.	$9 - 5 = 4$
3. Numerator: $4(3 - \sqrt{5})$.	$= 12 - 4\sqrt{5}$
4. Divide each term by 4.	$3 - \sqrt{5}$

***4*/(3 + √5) = 3 − √5. Check: (3 − √5)(3 + √5) = 4. ✓**

See it cancel, then use it

Learners meet the conjugate first as something they notice in the grid — the two surd cells cancelling — before it is named or used. Only then is it applied to denominators, with the reason (multiplying by 1) made explicit at each step. The proof stretch shows why the conjugate always works. The √ palette is on every surd answer box; the grid has its own √ and – keys.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Surds times surds: five products, including a single bracket.	~6 min
Movement	The surd expansion grid: fill cells, collect, notice cancelling. Stretch: prove $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$.	~15 min
Reflection	Clear the denominator: choose the multiplier, find the denominator, simplify.	~12 min
Creativity	Conjugate explorer: build pairs with products 1, 2 and a negative number.	~7 min
Rest	The two surd cells as mirror images.	~2 min
Nutrition	Intellectual: a trick that turns out to be a truth.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

$\sqrt{3} \times \sqrt{3} = 9$.
Ask: “What does $\sqrt{3}$ mean? What times itself gives 3?”

$(1 + \sqrt{2})^2 = 1 + 2$.
Ask: “Draw the grid for $(1 + \sqrt{2})(1 + \sqrt{2})$. How many cells?”

Multiplying only the denominator by $\sqrt{b}$.
Ask: “Is the value still the same? What are you multiplying the whole fraction by?”

Using the same bracket instead of the conjugate: $(3 + \sqrt{5})(3 + \sqrt{5})$.
Ask: “Do the surd cells cancel in that grid?”

Dividing only one term of the numerator by the denominator.
Ask: “Does every term of the numerator get divided?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: expanding brackets, and simplifying surds. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner multiplies $4/(3 + \sqrt{5})$ by $(3 + \sqrt{5})/(3 + \sqrt{5})$. The aim is a question that lets them see why it does not help.

- “What is the new denominator? Draw the grid.”
- “Is there still a surd?”

- “Which bracket would make the surd cells cancel?”
- “Why is it still fair to multiply by that?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is an answer like 12/4 – 4√5/4 accepted?
The value is recognised, but learners are encouraged to simplify fully (3 – √5).

Why rationalise at all, now that we have calculators?
It gives one agreed tidy form, so answers can be compared and collected easily. Historically it made hand calculation practical.

Is the proof required?
It is a stretch. The secure core is expanding with the grid and rationalising simple denominators.

What comes next?
Lesson 7: estimating and judging reasonableness — the end of the unit.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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